

Jeevan Kumar Kallempudi

Bluetooth SQA Engineer

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Career Objective:

A Bluetooth Software QA Engineer with 2.5+ years of experience in embedded software testing with good understanding of BT 5.2+EDR and profiles HFP, A2DP and have good understanding in the Networking, Wi-Fi and have good knowledge in Linux kernel. Seeking for a challenging & responsible position in an esteemed organization, which would be rewarding in terms of good learning experience and contribute to organizational and personal growth.

Key Skills:

Operating Systems --- Linux • RTOS • Zephyr • Android

Programming Languages --- C • Python

Software --- Jira • Ellisys • Com Probe Protocol Analyzer • Saleae LA • Picoscope • Audacity

Networking technology --- Bluetooth, Bluetooth Low Energy, BLE Audio, Auracast, Basics of Wi-Fi

Professional Experience:

Capgemini Engineering (In-House)

Project 1: LE Audio and AURACAST

Feb 2024 - Present

This project involves in creating POC for LE Audio and Auracast Feature.

- Prepared **AURACAST** POC w.r.t validation.
- Working with **nRF5340 Audio DK** from Nordic Semiconductors to validating **LE Audio** through the OTA logs.

Capgemini Engineering | NXP Semiconductors Pune (Client)

Project 1: RTOS Zephyr

Nov 2023 – Jan 2024

This project involves in BLE profile verification and validation with Zephyr OS.

- **Integration** of Zephyr OS includes **SDK cloning** and **building** from **GitHub**.
- **Upstream** profile execution and exploration.
- **Low Power Mode**, **IOP** Testing, **RvR Test** Execution.

Project 2: 3 – Wire UART (H5 without OOF control) Protocol Validation.

Sep 2023 - Nov 2023

This project involves in 3-Wire UART protocol (H5 without OOF control) verification and validation.

- Setup Configuration (CTS-RTS grounded).
- Bringing up 3rd party **H5 Protocol** supported chipsets (**RTL8822CS** (Fn-Link), **RTL8723BS**(Realtek)). Throughput comparison b/w NXP and 3rd party chipsets.
- Creation of Test plan and execution plan
- Validating H5 Protocol using HCI logs. Preparing Reports based on accurate testing results.

Project3: BT NXPUART Driver Validation

June 2023 – Sep 2023

This project involves in FW downloading using **NXPUART** driver for BT and **DEEP SLEEP** validation.

- Deep sleep setup configuration.
- Validation FW downloading and attaching at different baud rates.
- Deep sleep validation for NXP chipsets with **Logic Analyzer**.
- Stress test of **FW loading/unloading**, BT/BLE scan on/off by using **shell scripts**.

Project 4: NXP (iMX Host + SoC) Bluetooth Profile/Protocol Validation.

Oct 2022 - June 2023

This project involves in Testing of NXP controller with iMX host (**Linux**, **Android** 12, 13) for Bluetooth Technology.

- Involved in Manual (which includes Classic Bluetooth and BLE Profile and Protocols Validation) execution.
- Identifying and filing issues over **JIRA** tool.
- Analysing and debugging the issues with help of OTA logs, **btmon** and **btsnoop** logs.
- Validation of **BT/BLE AFH** channel classification in presence of external Wi-Fi traffic.
- Verification of multi-profile and stress test in congested environment.
- Creation of **semi - shell scripts** for ease of manual test execution.
- Validation of features like **Independent Reset** (In-Band, Out-of-Band) and **Suspend & Resume**.
- Involved in **BATS** (Controller Level Testing) and **WATS** (Profile Level Testing) Automation Execution.

Project 5: NXP RTOS Bluetooth Profile/Protocol Validation

Mar 2022 - Oct 2022

This project involves in Testing of Firecrest Controller and Host (**Ethermind Stack**) of NXP semiconductors on RTOS platform for Bluetooth Technology.

- Involved in Manual (which includes Classic Bluetooth and BLE Profile and Protocols Validation) execution.
- Creation of Test plan and execution plan.
- Identifying and reporting the bugs with L1 analysis of logs.
- Validation of **BT/BLE AFH** channel classification in presence of external Wi-Fi traffic.
- Verification of Stress test and multiple BT/BLE profiles.
- Design, development and test of **YAML scripts** for RTOS automation.
- Involved in Daily sanity execution (**CI/CD**) using **Jenkins**.

Education:

B. Tech REC, Visakhapatnam Electronics and Communication Engineering 8.38 CGPA	2018 – 2021
Diploma M.B.T.S Govt Polytechnic, Guntur Electronic and Communication Engineering 83.81%	2015 – 2018
SSC (10 th) Z.P. High School, Jalluru 9.2 CGPA	2014 - 2015